



```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
In [3]: df = pd.read_csv("C:/Users/rezay/Downloads/titanic_train.csv")
```

```
In [4]: df
```

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	210151
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	17566
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/OJ 310129
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373451
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112052
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W/ 6616
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376

891 rows × 12 columns

In [63]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass          891 non-null    int64
3   Name            891 non-null    object
4   Sex             891 non-null    object
5   Age            714 non-null    float64
6   SibSp          891 non-null    int64
7   Parch          891 non-null    int64
8   Ticket         891 non-null    object
9   Fare          891 non-null    float64
10  Cabin         204 non-null    object
11  Embarked       889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

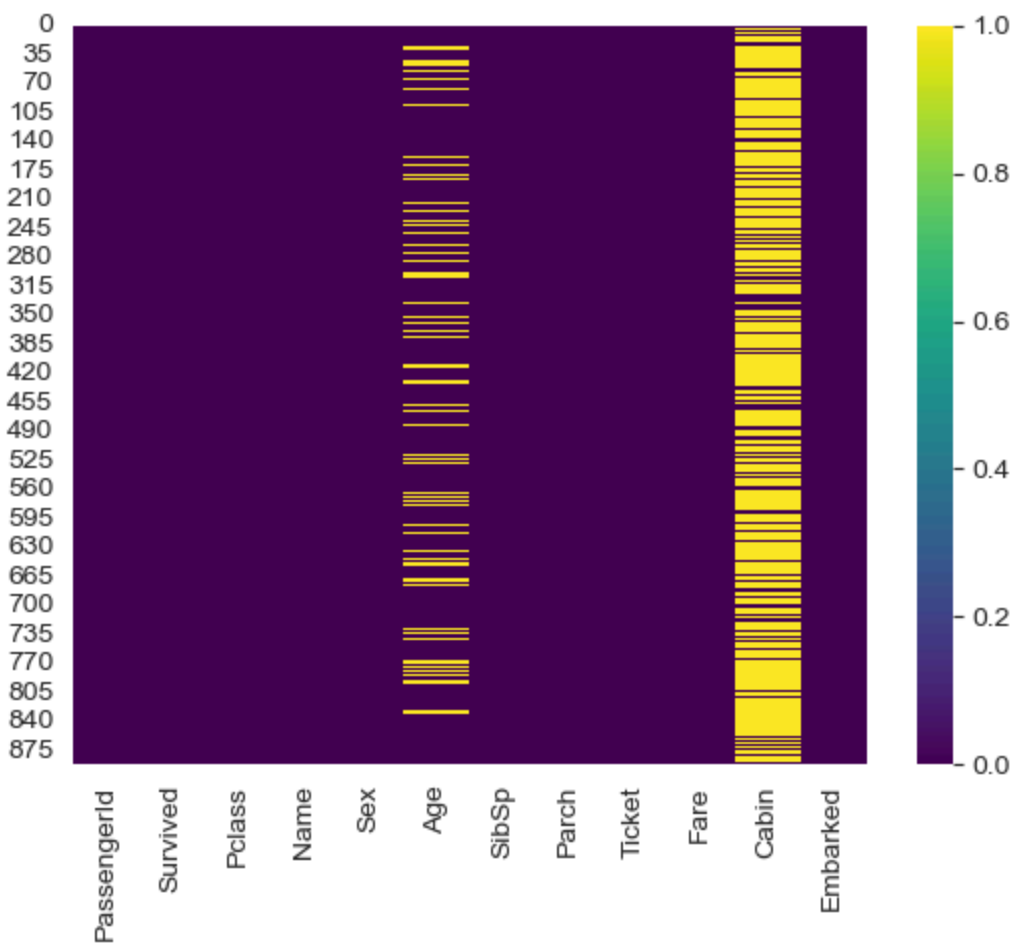
In [64]: `df.describe()`

Out[64]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

In [65]: `sns.heatmap(df.isnull(),cmap='viridis')`

Out[65]: `<Axes: >`

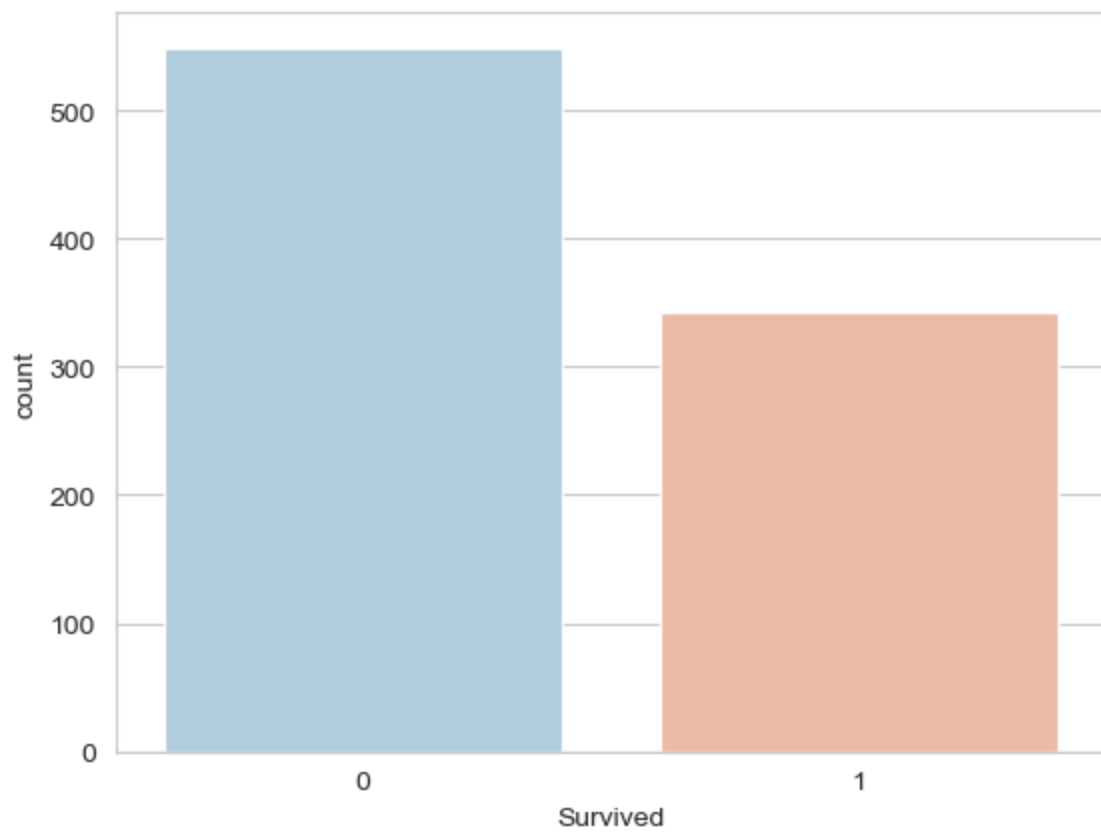


```
In [66]: sns.set_style('whitegrid')
sns.countplot(x='Survived',data=df,palette='RdBu_r')
```

C:\Users\rezay\AppData\Local\Temp\ipykernel_16392\2271557996.py:2: FutureWarning:

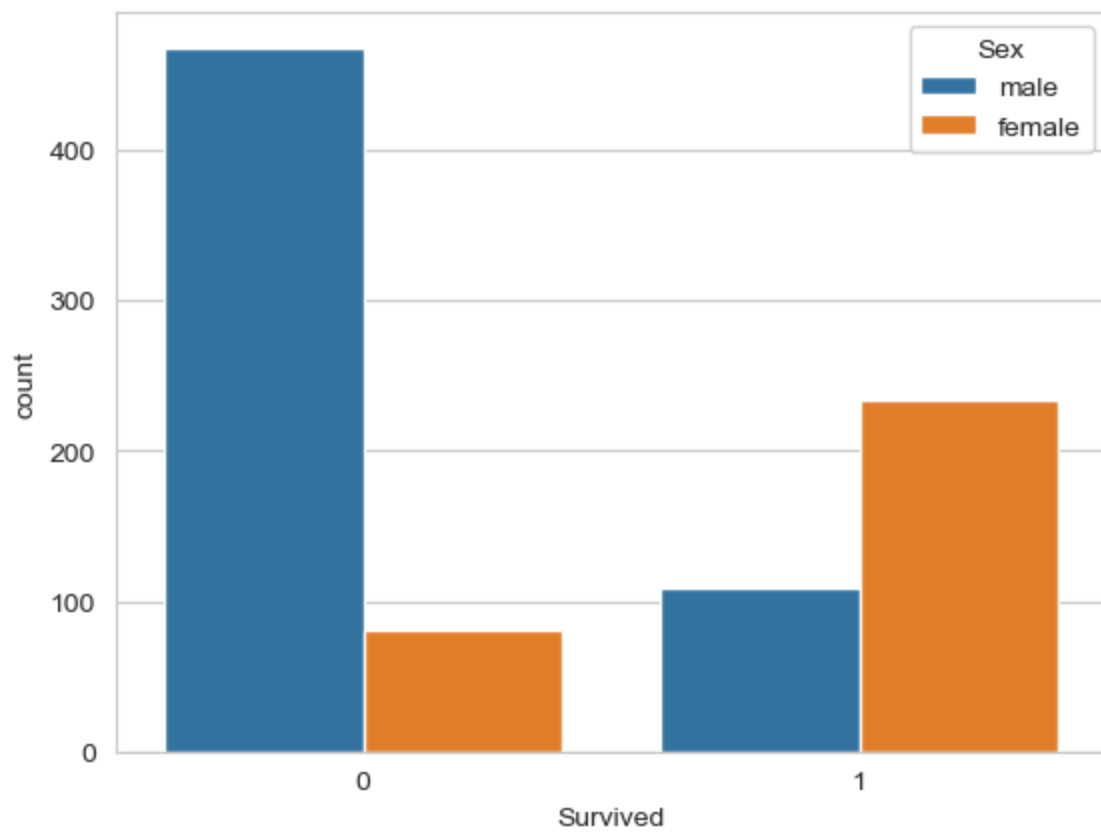
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
Out[66]: <Axes: xlabel='Survived', ylabel='count'>
```



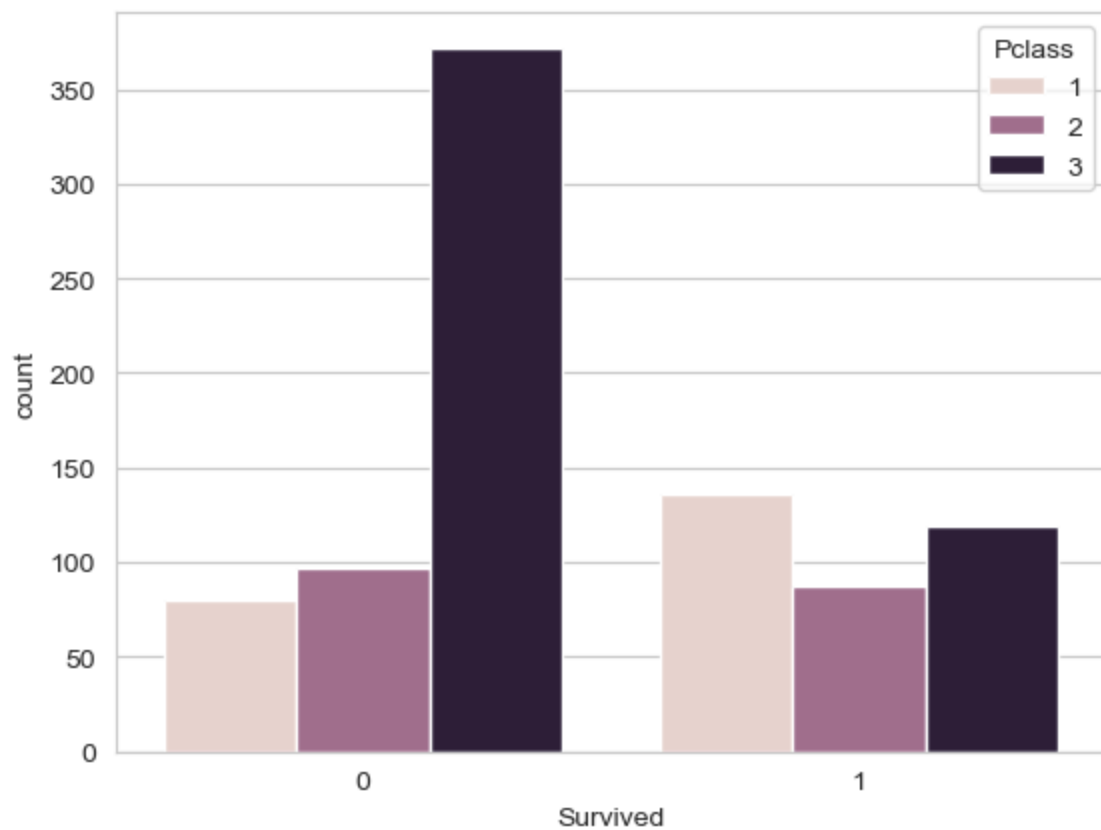
```
In [67]: sns.countplot(x='Survived',hue='Sex',data=df)
```

```
Out[67]: <Axes: xlabel='Survived', ylabel='count'>
```



```
In [68]: sns.countplot(x='Survived',hue='Pclass',data=df)
```

```
Out[68]: <Axes: xlabel='Survived', ylabel='count'>
```



```
In [69]: sns.distplot(df['Age'].dropna(), kde = False, bins=20)
```

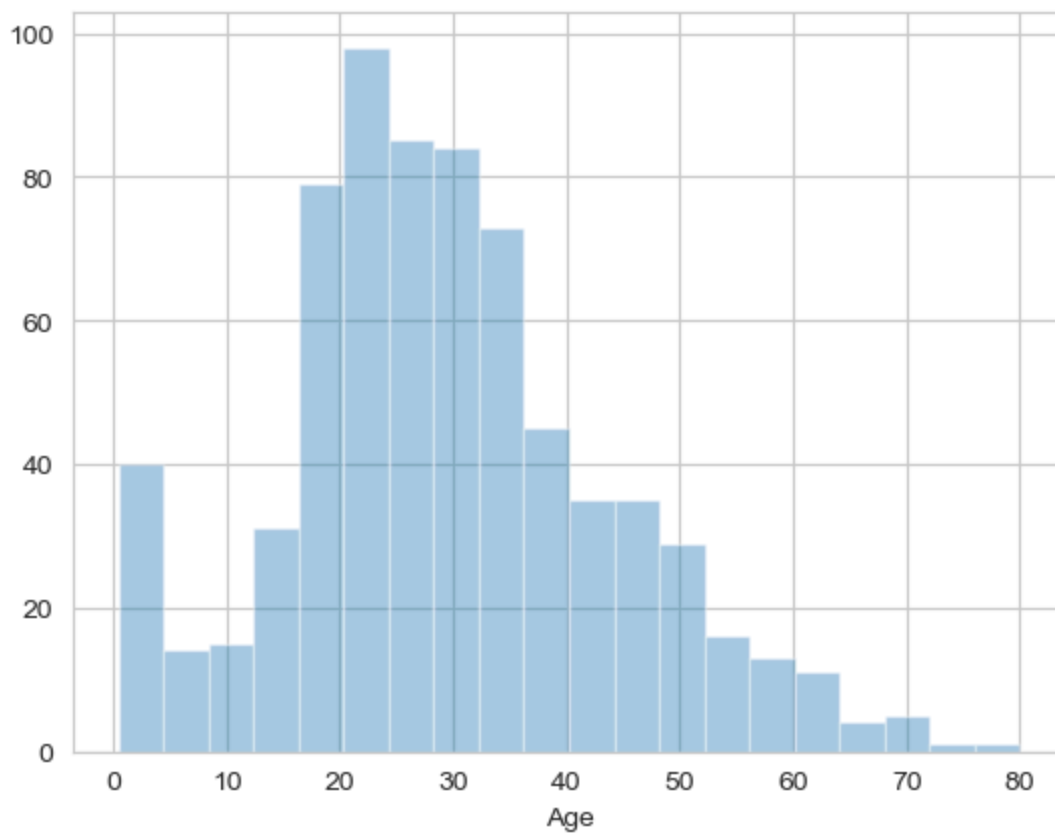
C:\Users\rezay\AppData\Local\Temp\ipykernel_16392\3408280264.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

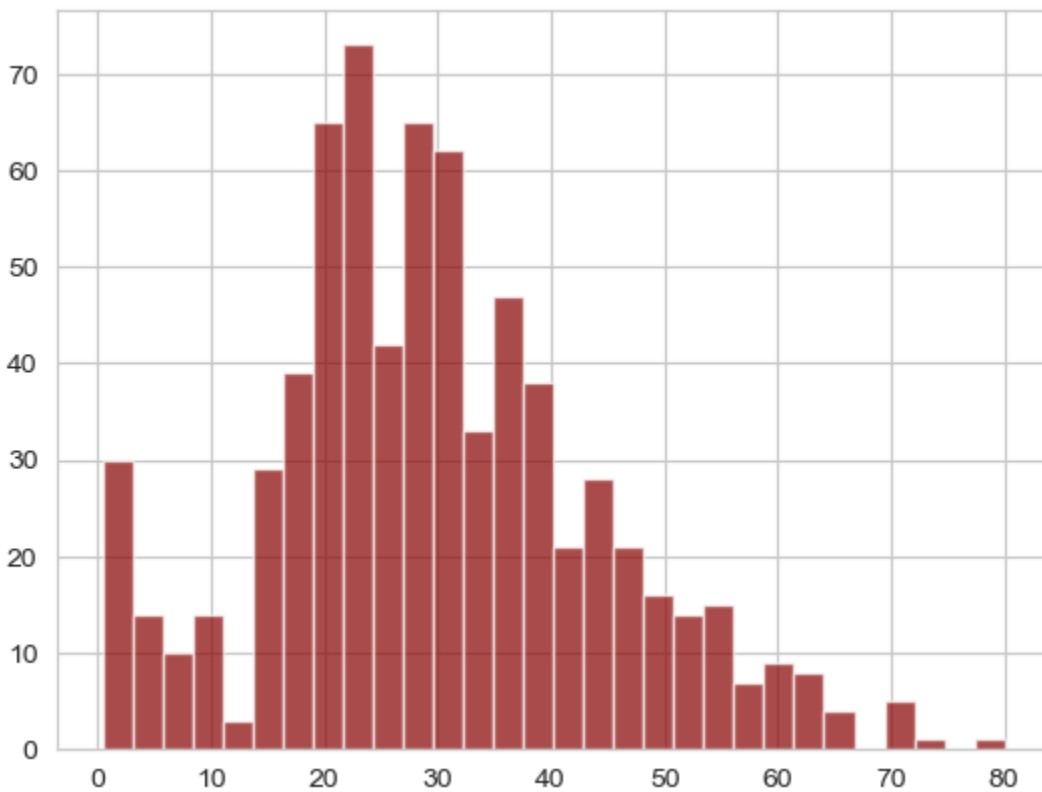
For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
Out[69]: <Axes: xlabel='Age'>
```



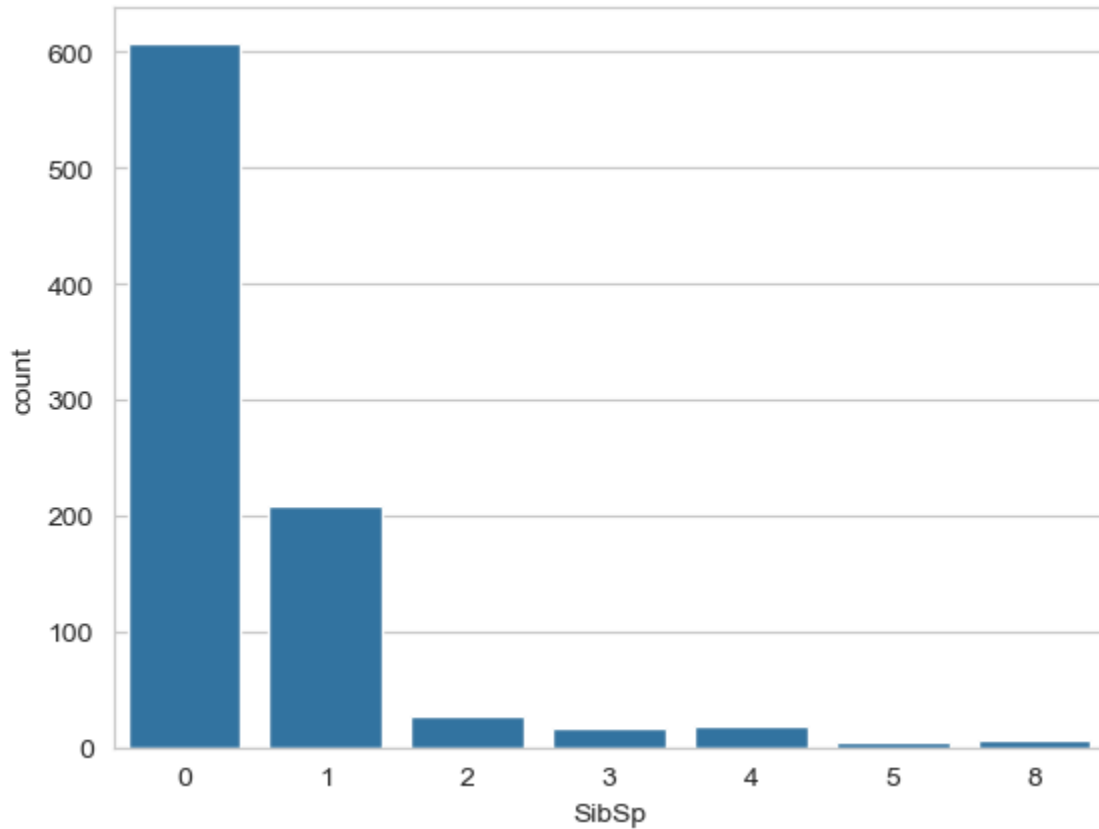
```
In [70]: df['Age'].hist(bins=30,color='darkred',alpha=0.7)
```

```
Out[70]: <Axes: >
```




```
In [71]: sns.countplot(x='SibSp',data=df)
```

```
Out[71]: <Axes: xlabel='SibSp', ylabel='count'>
```

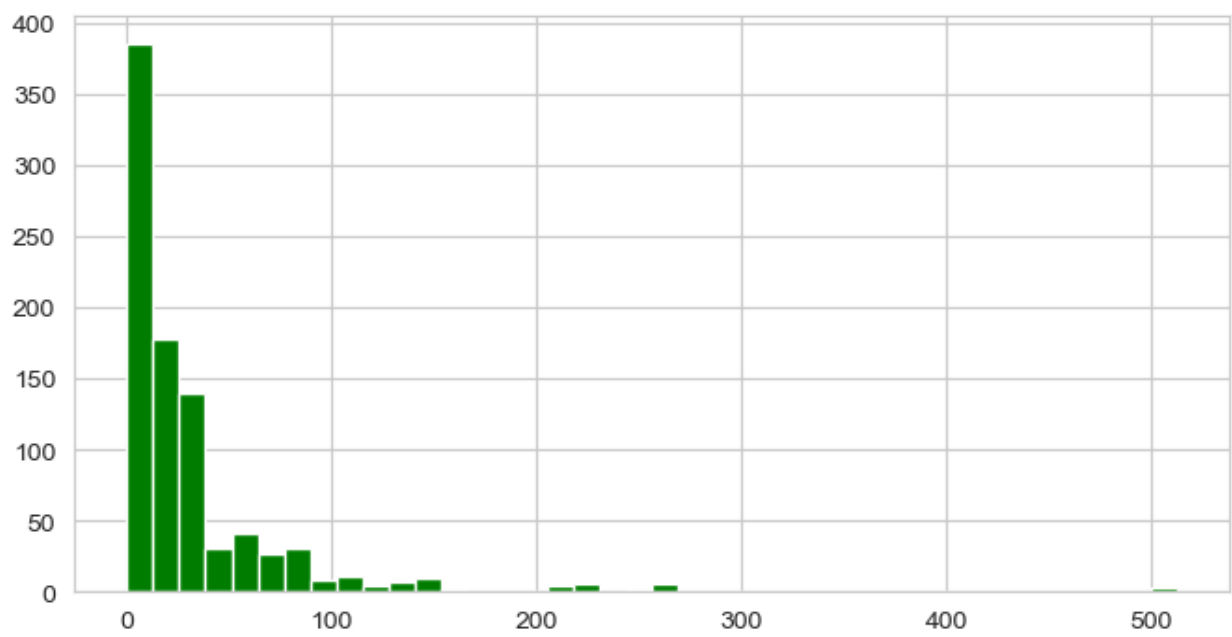


```
In [72]: df.columns
```

```
Out[72]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',  
              'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],  
              dtype='object')
```

```
In [73]: df['Fare'].hist(color='green',bins=40,figsize=(8,4))
```

```
Out[73]: <Axes: >
```



In [74]: `pip install cufflinks`

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: cufflinks in c:\users\rezay\appdata\roaming\python\python312\site-packages (0.17.3)
Requirement already satisfied: numpy>=1.9.2 in c:\programdata\anaconda3\lib\site-packages (from cufflinks) (1.26.4)
Requirement already satisfied: pandas>=0.19.2 in c:\programdata\anaconda3\lib\site-packages (from cufflinks) (2.2.2)
Requirement already satisfied: plotly>=4.1.1 in c:\programdata\anaconda3\lib\site-packages (from cufflinks) (5.22.0)
Requirement already satisfied: six>=1.9.0 in c:\programdata\anaconda3\lib\site-packages (from cufflinks) (1.16.0)
Requirement already satisfied: colorlover>=0.2.1 in c:\users\rezay\appdata\roaming\python\python312\site-packages (from cufflinks) (0.3.0)
Requirement already satisfied: setuptools>=34.4.1 in c:\programdata\anaconda3\lib\site-packages (from cufflinks) (69.5.1)
Requirement already satisfied: ipython>=5.3.0 in c:\programdata\anaconda3\lib\site-packages (from cufflinks) (8.25.0)
Requirement already satisfied: ipywidgets>=7.0.0 in c:\programdata\anaconda3\lib\site-packages (from cufflinks) (7.8.1)
Requirement already satisfied: decorator in c:\programdata\anaconda3\lib\site-packages (from ipython>=5.3.0->cufflinks) (5.1.1)
Requirement already satisfied: jedi>=0.16 in c:\programdata\anaconda3\lib\site-packages (from ipython>=5.3.0->cufflinks) (0.18.1)
Requirement already satisfied: matplotlib-inline in c:\programdata\anaconda3\lib\site-packages (from ipython>=5.3.0->cufflinks) (0.1.6)
Requirement already satisfied: prompt-toolkit<3.1.0,>=3.0.41 in c:\programdata\anaconda3\lib\site-packages (from ipython>=5.3.0->cufflinks) (3.0.43)
Requirement already satisfied: pygments>=2.4.0 in c:\programdata\anaconda3\lib\site-packages (from ipython>=5.3.0->cufflinks) (2.15.1)
Requirement already satisfied: stack-data in c:\programdata\anaconda3\lib\site-packages (from ipython>=5.3.0->cufflinks) (0.2.0)
Requirement already satisfied: traitlets>=5.13.0 in c:\programdata\anaconda3\lib\site-packages (from ipython>=5.3.0->cufflinks) (5.14.3)
Requirement already satisfied: colorama in c:\programdata\anaconda3\lib\site-packages (from ipython>=5.3.0->cufflinks) (0.4.6)
Requirement already satisfied: comm>=0.1.3 in c:\programdata\anaconda3\lib\site-packages (from ipywidgets>=7.0.0->cufflinks) (0.2.1)
Requirement already satisfied: ipython-genutils<=0.2.0 in c:\programdata\anaconda3\lib\site-packages (from ipywidgets>=7.0.0->cufflinks) (0.2.0)
Requirement already satisfied: widgetsnbextension<=3.6.6 in c:\programdata\anaconda3\lib\site-packages (from ipywidgets>=7.0.0->cufflinks) (3.6.6)
Requirement already satisfied: jupyterlab-widgets<3,>=1.0.0 in c:\programdata\anaconda3\lib\site-packages (from ipywidgets>=7.0.0->cufflinks) (1.0.0)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\anaconda3\lib\site-packages (from pandas>=0.19.2->cufflinks) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas>=0.19.2->cufflinks) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3\lib\site-packages (from pandas>=0.19.2->cufflinks) (2023.3)
Requirement already satisfied: tenacity>=6.2.0 in c:\programdata\anaconda3\lib\site-packages (from plotly>=4.1.1->cufflinks) (8.2.2)
Requirement already satisfied: packaging in c:\programdata\anaconda3\lib\site-packages (from plotly>=4.1.1->cufflinks) (23.2)
Requirement already satisfied: parso<0.9.0,>=0.8.0 in c:\programdata\anaconda3\

lib\site-packages (from jedi>=0.16->ipython>=5.3.0->cufflinks) (0.8.3)
Requirement already satisfied: wcwidth in c:\programdata\anaconda3\lib\site-packages (from prompt-toolkit<3.1.0,>=3.0.41->ipython>=5.3.0->cufflinks) (0.2.5)
Requirement already satisfied: notebook>=4.4.1 in c:\programdata\anaconda3\lib\site-packages (from widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (7.0.8)
Requirement already satisfied: executing in c:\programdata\anaconda3\lib\site-packages (from stack-data->ipython>=5.3.0->cufflinks) (0.8.3)
Requirement already satisfied: asttokens in c:\programdata\anaconda3\lib\site-packages (from stack-data->ipython>=5.3.0->cufflinks) (2.0.5)
Requirement already satisfied: pure-eval in c:\programdata\anaconda3\lib\site-packages (from stack-data->ipython>=5.3.0->cufflinks) (0.2.2)
Requirement already satisfied: jupyter-server<3,>=2.4.0 in c:\programdata\anaconda3\lib\site-packages (from notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.14.1)
Requirement already satisfied: jupyterlab-server<3,>=2.22.1 in c:\programdata\anaconda3\lib\site-packages (from notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.25.1)
Requirement already satisfied: jupyterlab<4.1,>=4.0.2 in c:\programdata\anaconda3\lib\site-packages (from notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (4.0.11)
Requirement already satisfied: notebook-shim<0.3,>=0.2 in c:\programdata\anaconda3\lib\site-packages (from notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.2.3)
Requirement already satisfied: tornado>=6.2.0 in c:\programdata\anaconda3\lib\site-packages (from notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (6.4.1)
Requirement already satisfied: anyio>=3.1.0 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (4.2.0)
Requirement already satisfied: argon2-cffi>=21.1 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (21.3.0)
Requirement already satisfied: jinja2>=3.0.3 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (3.1.4)
Requirement already satisfied: jupyter-client>=7.4.4 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (8.6.0)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (5.7.2)
Requirement already satisfied: jupyter-events>=0.9.0 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.10.0)
Requirement already satisfied: jupyter-server-terminals>=0.4.4 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.4.4)
Requirement already satisfied: nbconvert>=6.4.4 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (7.10.0)
Requirement already satisfied: nbformat>=5.3.0 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (5.9.2)

Requirement already satisfied: overrides>=5.0 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (7.4.0)

Requirement already satisfied: prometheus-client>=0.9 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (0.14.1)

Requirement already satisfied: pywinpty>=2.0.1 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (2.0.10)

Requirement already satisfied: pyzmq>=24 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (25.1.2)

Requirement already satisfied: send2trash>=1.8.2 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (1.8.2)

Requirement already satisfied: terminado>=0.8.3 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (0.17.1)

Requirement already satisfied: websocket-client>=1.7 in c:\programdata\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (1.8.0)

Requirement already satisfied: async-lru>=1.0.0 in c:\programdata\anaconda3\lib\site-packages (from jupyterlab<4.1,>=4.0.2->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (2.0.4)

Requirement already satisfied: ipykernel in c:\programdata\anaconda3\lib\site-packages (from jupyterlab<4.1,>=4.0.2->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (6.28.0)

Requirement already satisfied: jupyter-lsp>=2.0.0 in c:\programdata\anaconda3\lib\site-packages (from jupyterlab<4.1,>=4.0.2->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (2.2.0)

Requirement already satisfied: babel>=2.10 in c:\programdata\anaconda3\lib\site-packages (from jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (2.11.0)

Requirement already satisfied: json5>=0.9.0 in c:\programdata\anaconda3\lib\site-packages (from jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (0.9.6)

Requirement already satisfied: jsonschema>=4.18.0 in c:\programdata\anaconda3\lib\site-packages (from jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (4.19.2)

Requirement already satisfied: requests>=2.31 in c:\programdata\anaconda3\lib\site-packages (from jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (2.32.2)

Requirement already satisfied: idna>=2.8 in c:\programdata\anaconda3\lib\site-packages (from anyio>=3.1.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (3.7)

Requirement already satisfied: sniffio>=1.1 in c:\programdata\anaconda3\lib\site-packages (from anyio>=3.1.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (1.3.0)

Requirement already satisfied: argon2-cffi-bindings in c:\programdata\anaconda3\lib\site-packages (from argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (21.2.0)

Requirement already satisfied: MarkupSafe>=2.0 in c:\programdata\anaconda3\lib\site-packages (from jinja2>=3.0.3->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~3.6.6->ipywidgets>=7.0.0->cufflinks) (2.1.3)

Requirement already satisfied: attrs>=22.2.0 in c:\programdata\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (23.1.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in c:\programdata\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2023.7.1)

Requirement already satisfied: referencing>=0.28.4 in c:\programdata\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.30.2)

Requirement already satisfied: rpds-py>=0.7.1 in c:\programdata\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.10.6)

Requirement already satisfied: platformdirs>=2.5 in c:\programdata\anaconda3\lib\site-packages (from jupyter-core!=5.0.*,>=4.12->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (3.10.0)

Requirement already satisfied: pywin32>=300 in c:\programdata\anaconda3\lib\site-packages (from jupyter-core!=5.0.*,>=4.12->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (305.1)

Requirement already satisfied: python-json-logger>=2.0.4 in c:\programdata\anaconda3\lib\site-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.0.7)

Requirement already satisfied: pyyaml>=5.3 in c:\programdata\anaconda3\lib\site-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (6.0.1)

Requirement already satisfied: rfc3339-validator in c:\programdata\anaconda3\lib\site-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.1.4)

Requirement already satisfied: rfc3986-validator>=0.1.1 in c:\programdata\anaconda3\lib\site-packages (from jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.1.1)

Requirement already satisfied: beautifulsoup4 in c:\programdata\anaconda3\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (4.12.3)

Requirement already satisfied: bleach!=5.0.0 in c:\programdata\anaconda3\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (4.1.0)

Requirement already satisfied: defusedxml in c:\programdata\anaconda3\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.7.1)

Requirement already satisfied: jupyterlab-pygments in c:\programdata\anaconda3\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.1.2)

Requirement already satisfied: mistune<4,>=2.0.3 in c:\programdata\anaconda3\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.0.4)

Requirement already satisfied: nbclient>=0.5.0 in c:\programdata\anaconda3\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.8.0)

Requirement already satisfied: pandocfilters>=1.4.1 in c:\programdata\anaconda3\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (1.5.0)

Requirement already satisfied: tinycss2 in c:\programdata\anaconda3\lib\site-packages (from nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (1.2.1)

Requirement already satisfied: fastjsonschema in c:\programdata\anaconda3\lib\site-packages (from nbformat>=5.3.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.16.2)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.31->jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.0.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.31->jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.2.2)

Requirement already satisfied: certifi>=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests>=2.31->jupyterlab-server<3,>=2.22.1->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2024.6.2)

Requirement already satisfied: debugpy>=1.6.5 in c:\programdata\anaconda3\lib\site-packages (from ipykernel->jupyterlab<4.1,>=4.0.2->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (1.6.7)

Requirement already satisfied: nest-asyncio in c:\programdata\anaconda3\lib\site-packages (from ipykernel->jupyterlab<4.1,>=4.0.2->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (1.6.0)

Requirement already satisfied: psutil in c:\programdata\anaconda3\lib\site-packages (from ipykernel->jupyterlab<4.1,>=4.0.2->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (5.9.0)

Requirement already satisfied: webencodings in c:\programdata\anaconda3\lib\site-packages (from bleach!=5.0.0->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (0.5.1)

Requirement already satisfied: fqdn in c:\users\rezay\appdata\roaming\python\python312\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-event-s>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (1.5.1)

Requirement already satisfied: isoduration in c:\users\rezay\appdata\roaming\python\python312\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (20.11.0)

Requirement already satisfied: jsonpointer>1.13 in c:\programdata\anaconda3\lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-event-s>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.1)

Requirement already satisfied: uri-template in c:\users\rezay\appdata\roaming\python\python312\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (1.3.0)

Requirement already satisfied: webcolors>=1.11 in c:\users\rezay\appdata\roaming\python\python312\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (24.11.1)

Requirement already satisfied: cffi>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (1.16.0)

Requirement already satisfied: soupsieve>1.2 in c:\programdata\anaconda3\lib\site-packages (from beautifulsoup4->nbconvert>=6.4.4->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.5)

Requirement already satisfied: pycparser in c:\programdata\anaconda3\lib\site-packages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (2.21)

Requirement already satisfied: arrow>=0.15.0 in c:\programdata\anaconda3\lib\site-packages (from isoduration->jschema[format-nongpl]>=4.18.0->jupyter-event>=0.9.0->jupyter-server<3,>=2.4.0->notebook>=4.4.1->widgetsnbextension~=3.6.6->ipywidgets>=7.0.0->cufflinks) (1.2.3)

Note: you may need to restart the kernel to use updated packages.

```
In [75]: import cufflinks as cf
         cf.go_offline()
```

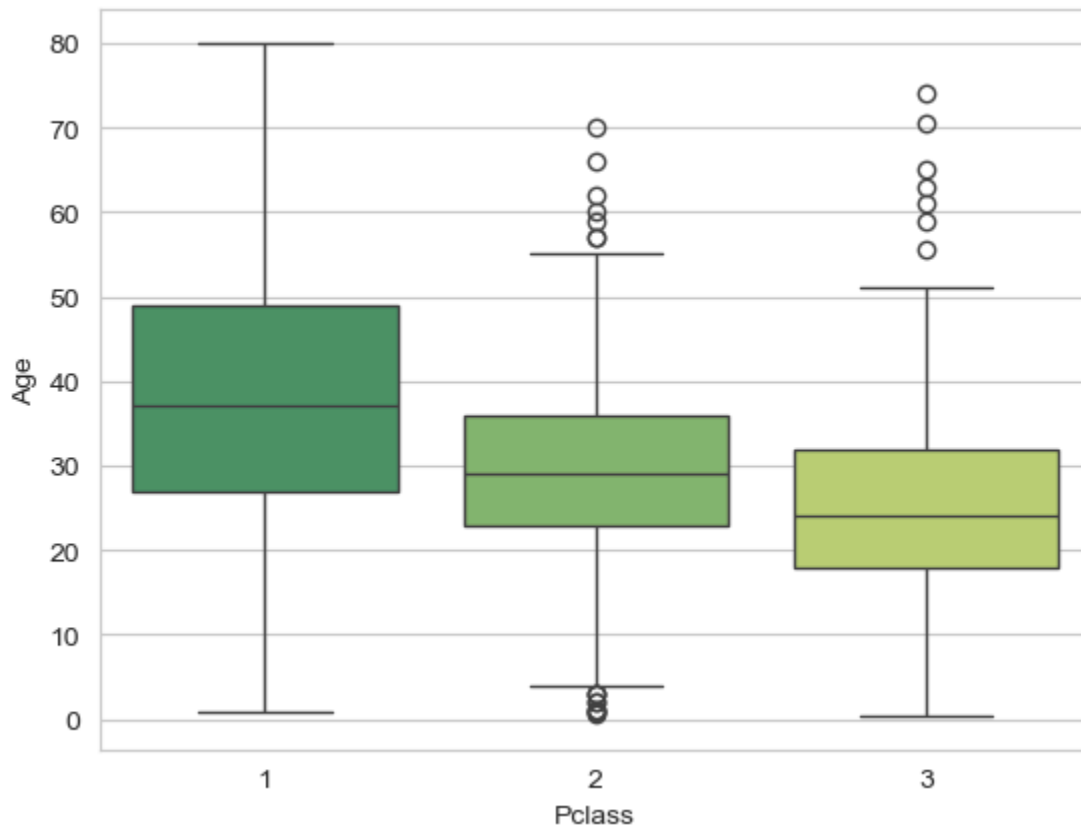
```
In [76]: df['Fare'].iplot(kind='hist',bins=30,color='green')
```

```
In [77]: sns.boxplot(x='Pclass',y='Age',data=df,palette='summer')
```


C:\Users\rezay\AppData\Local\Temp\ipykernel_16392\4028820417.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

Out[77]: <Axes: xlabel='Pclass', ylabel='Age'>



```
In [78]: def impute_age(cols):  
    Age = cols[0]  
    Pclass = cols[1]  
    if pd.isnull(Age):  
        if Pclass == 1:  
            return 37  
        elif Pclass == 2:  
            return 29  
        else:  
            return 24  
    else:  
        return Age
```

```
In [79]: df['Age'] = df[['Age', 'Pclass']].apply(impute_age, axis=1)
```

```
C:\Users\rezay\AppData\Local\Temp\ipykernel_16392\3983263054.py:2: FutureWarning:
```

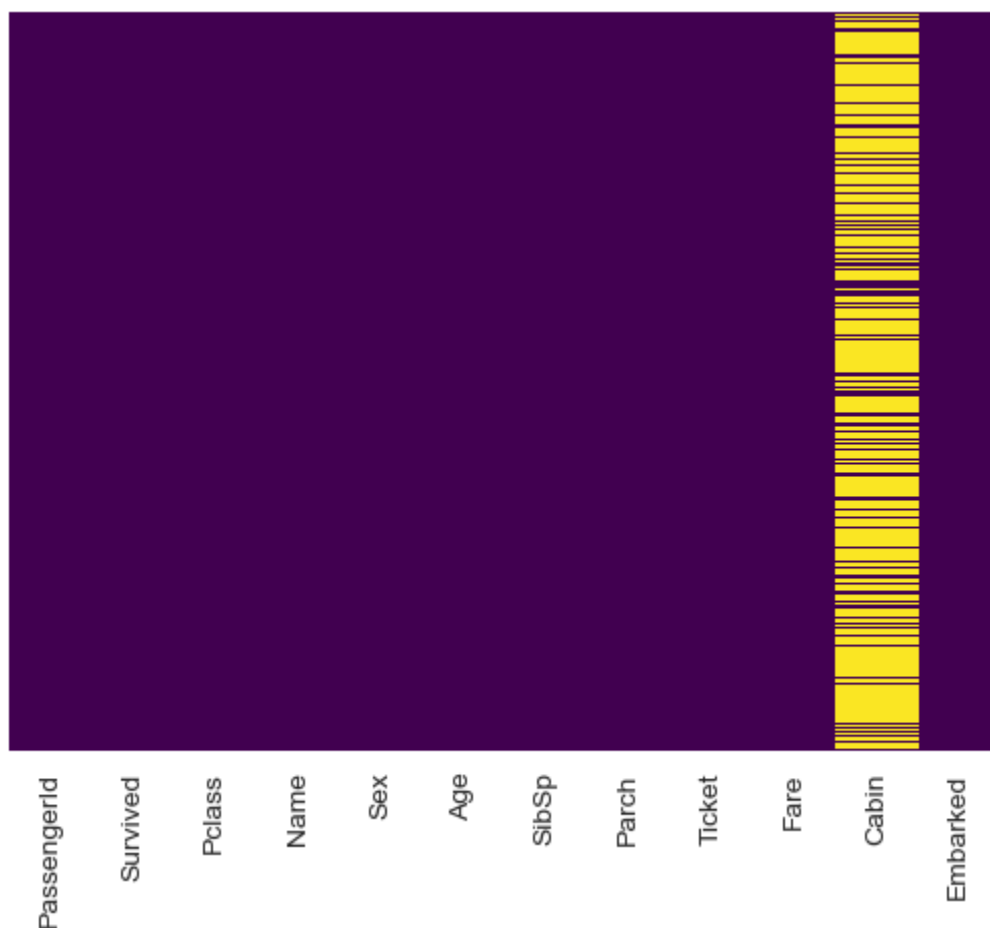
```
Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame behavior). To access a value by position, use `ser.iloc[pos]`
```

```
C:\Users\rezay\AppData\Local\Temp\ipykernel_16392\3983263054.py:3: FutureWarning:
```

```
Series.__getitem__ treating keys as positions is deprecated. In a future version, integer keys will always be treated as labels (consistent with DataFrame behavior). To access a value by position, use `ser.iloc[pos]`
```

```
In [80]: sns.heatmap(df.isnull(),yticklabels=False,cbar=False,cmap='viridis')
```

```
Out[80]: <Axes: >
```



```
In [81]: df.drop('Cabin',axis=1,inplace=True)
```

```
In [82]: df.head()
```

```
Out[82]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

```
In [83]: df.isnull().sum()
```

```
Out[83]: PassengerId    0
Survived      0
Pclass        0
Name          0
Sex           0
Age           0
SibSp         0
Parch         0
Ticket        0
Fare          0
Embarked      2
dtype: int64
```

```
In [84]: df.dropna(inplace=True)
```

```
In [85]: df.isnull().sum()
```

```
Out[85]: PassengerId    0
         Survived      0
         Pclass        0
         Name          0
         Sex           0
         Age           0
         SibSp         0
         Parch         0
         Ticket        0
         Fare          0
         Embarked      0
         dtype: int64
```

```
In [86]: df.isnull().sum()
```

```
Out[86]: PassengerId    0
         Survived      0
         Pclass        0
         Name          0
         Sex           0
         Age           0
         SibSp         0
         Parch         0
         Ticket        0
         Fare          0
         Embarked      0
         dtype: int64
```

```
In [87]: sex = pd.get_dummies(df['Sex'],drop_first=True)
         embark = pd.get_dummies(df['Embarked'],drop_first=True)
```

```
In [88]: df
```

Out[88]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	210151
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	17566
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/OJ 310129
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373451
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112052
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	24.0	1	2	W 66153
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376

889 rows x 11 columns

In [89]: `df = pd.concat([df,sex,embark],axis=1)`

In [90]: `df`

Out[90]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STC 31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3734
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	2115
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	1120
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	24.0	1	2	W 66
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	1113
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	3703

889 rows × 14 columns

In [91]: `df['male'] = df['male'].astype(int)`

In [92]: `df['Q'] = df['Q'].astype(int)`

```
In [93]: df['S'] = df['S'].astype(int)
```

```
In [94]: df
```

```
Out[94]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	2101517
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	1759662
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/OJ 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138033
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373451
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	2115365
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	1120529
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	24.0	1	2	W 661636
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	1113698
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	3703763

889 rows × 14 columns

```
In [95]: from sklearn.model_selection import train_test_split
```

```
X = df[['PassengerId', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare', 'male', 'Q', 'S']]
y = df['Survived']
```

```
In [96]: X_train , X_test , y_train , y_test = train_test_split(X,y,test_size=0.33,rand
```

```
In [97]: from sklearn.linear_model import LogisticRegression
```

```
In [98]: logmodel = LogisticRegression()
logmodel.fit(X_train,y_train)
```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\linear_model_logistic.py:469: ConvergenceWarning:

lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
<https://scikit-learn.org/stable/modules/preprocessing.html>
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
Out[98]: LogisticRegression
```

```
LogisticRegression()
```

```
In [99]: predictions = logmodel.predict(X_test)
```

```
In [100]: from sklearn.metrics import classification_report
```

```
In [101]: print(classification_report(y_test,predictions))
```

	precision	recall	f1-score	support
0	0.83	0.86	0.84	184
1	0.75	0.70	0.72	110
accuracy			0.80	294
macro avg	0.79	0.78	0.78	294
weighted avg	0.80	0.80	0.80	294